

Exercises

Level - 1

(Problems Based on Fundamentals)

1. Find the maximum and the minimum values of each of the following, functions (without using derivative)

- (i) $f(x) = x^2 - 4x + 10$
- (ii) $f(x) = x^2 + 6x - 15$
- (iii) $f(x) = (x - 3)^2 + 5$
- (iv) $f(x) = -(x - 1)^2 + 2$
- (v) $f(x) = |x + 2|$
- (vi) $f(x) = -|x - 1| + 5$
- (vii) $f(x) = |x - 1| + |x - 3|$
- (viii) $f(x) = |x + 2| + |x - 3|$
- (ix) $f(x) = x^3 - 1$
- (x) $f(x) = 3 \sin x + 4 \cos x + 10$
- (xi) $f(x) = \quad + \cos x$
- (xi) $f(x) = \sin x - \cos x$
- (xii) $f(x) = \sin(2x) + 5$
- (xiii) $f(x) = 3 - 2 \sin x$
- (xiv) $f(x) = 2 \cos x + 3$
- (xv) $f(x) = 3 - 4 \cos x$
- (xvi) $f(x) = 2 \sin^2 x + 4$
- (xvii) $f(x) = 5 - 3 \sin^2 x$
- (xviii) $f(x) = 3 \sin^2 x + 2 \cos^2 x$
- (xix) $f(x) = 4 \sin^2 x + 5 \cos^2 x + 6 \sin x \cos x +$

2. Find the maximum and the minimum values of each of the following function without using derivative

- (i) $f(x) = \sin(\sin x)$
- (ii) $f(x) = \cos(\cos x)$

(iii) $f(x) = \sin(\sin x) + \cos(\sin x)$

(iv) $f(x) = \sin^2 x + \cos^4 x$

(v) $f(x) = \sin^4 x + \cos^2 x$

(vi) $f(x) = \sin^4 x + \cos^4 x$

(vii) $f(x) = \sin^6 x + \cos^6 x$

(viii) $f(x) = \sin^2(\sin x) + \cos^2(\cos x)$

(ix) $f(x) = |\sin x| + |\cos x|$

(x) $f(x) = \sin x + \operatorname{cosec} x$ for all x in $\left(0, \frac{\pi}{2}\right)$

3. Find the min value of $f(x) = |x - a| + |x - b|$ where $0 < a < b$.

4. Find the min value of $f(x) = |x - a| + |x - b| + |x - c|$ where $0 < a < b < c$.

5. Let $f(x) = |x - a| + |x - b| + |x - c| + |x - d|$ where $a < b < c < d$, find the minimum value of $f(x)$.

Local Max or Local Min

6. Find the points of extremum of the function

$$f(x) = (x - 1)^2(x - 2)^3$$

7. Find the points of extremum of the function

$$f(x) = (x - 2)x^{2/3}$$

8. Find the points of extremum of the function

$$f(x) = x^3 - 6x^2 + 12x - 8.$$

9. Find the points of extremum of the function $f(x)$ for which

$$f'(x) = (x - 1)(x - 2)^2(x - 3)^3(x - 4)^4$$

10. Find the points of extremum of the function

$$f(x) = \sqrt{2x^2 - x + 2}.$$

11. Find the points of extremum of the function

$$f(x) = x - \ln(1 + x^2).$$

12. Discuss the extremum of

$$f(x) = \begin{cases} 1 + \sin x & : x < 0 \\ x^2 - x + 1 & : x \geq 0 \end{cases}$$

13. Find the points of extremum of the function

$$f(x) = 1 + (x - 1)^{2/3}$$

14. Find the points of extremum of the function

$$f(x) = 1 + (x - 2)^{4/5}$$

15. Find the points of extremum of the function

$$f(x) = \begin{cases} x^2 & : x \leq 0 \\ 2\sin x & : x > 0 \end{cases}$$

16. Find the points of extremum of the function

$$f(x) = |x| + |x^2 - 1|.$$

17. Find the points of extremum of the function

$$f(x) = 2\log(x - 2) - x^2 + 4x + 1.$$

Max or Min at the end points

18. Find the absolute max and min of

$$f(x) = x^3 - 3x^2 + 1, \text{ for all } x \text{ in } \left[-\frac{1}{2}, 4\right]$$

19. Find the absolute max or min of

$$f(x) = x^{2/3}(5 - 2x) \text{ in } [-1, 2]$$

20. Find the absolute max or min of

$$f(x) = x + 2\sqrt{x}, \forall x \in [0, 4]$$

21. Find the absolute max or min of

$$f(x) = x^2 \ln x, \forall x \in [1, e]$$

22. Find the absolute max or min of

$$f(x) = x + \sin 2x, \forall x \in [0, 2\pi]$$

23. Find the absolute max or min of

$$f(x) = \sin x + \frac{1}{2} \cos 2x, \forall x \in \left[0, \frac{\pi}{2}\right]$$

24. Find the max or min values of

$$f(x) = \tan^{-1} x - \frac{1}{2} \ln x, \forall x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$$

25. Find the max or min values of

$$f(x) = \{(1 - x^2)(2x^2 + 1)\}^{1/2}, \forall x \in [-1, 1]$$

26. Find the greatest and least values of

$$f(x) = \frac{\sin 2x}{\sin\left(x + \frac{\pi}{4}\right)} \text{ in } \left[0, \frac{\pi}{2}\right]$$

27. Find the greatest and least values of

$$f(x) = \cos x + \cos(\sqrt{2}x)$$

28. Find the greatest and least value of the function

$$f(x) = \frac{1}{\sin x + 4} + \frac{1}{\cos x - 4} \text{ for all } x \text{ in } R.$$

29. Find the global max or global min of

$$f(x) = \frac{14}{x^4 - 8x^2 + 2} \text{ for all } x \text{ in } R.$$

30. Find the global max or global min of

$$f(x) = (x - 1)^2 \sqrt{x^2 - 2x + 3} \text{ in } [0, 3]$$

31. Find the max and min values of the function

$$y = \frac{x^2 - 7x + 6}{x - 10}$$

32. Find the max or min values of

$$f(x) = 2\sin x + \cos 2x$$

33. Find the max or min values of

$$f(x) = 3x^4 - 2x^3 - 3x^2 + 10$$

34. Find the max or min values of

$$f(x) = \ln(x^4 - 2x^2 + 3)$$

35. Find the max or min values of

$$f(x) = \sqrt{3x^2 - 2x^3}$$

36. Find the max or min values of

$$f(x) = \frac{10}{3x^4 + 4x^3 - 12x^2 + 11}$$

37. Find the max or min values of

$$f(x) = x^5 + 5x^4 + 5x^3 - 1$$

Mensuration Problems

38. Show that of all the rectangles with a given perimeter, the square has the largest area.
39. Show that of all the rectangles inscribed in a given circle, the square has the maximum area.
40. If the sum of the lengths of the hypotenuse and a side of a right angled triangle is given, show that the area of the triangle is max when the angle between them is $\frac{\pi}{3}$.
41. Show that the triangle of maximum area that can be inscribed in a given circle is an equilateral triangle.
42. Show that the semi vertical angle of a cone of max volume and of given slant height is $\tan^{-1}(\sqrt{2})$.
43. Show that the semi vertical angle of a right circular cone of given surface area and max volume is $\sin^{-1}\left(\frac{1}{3}\right)$.
44. Show that the height of the cylinder of max volume that can be inscribed in a sphere of radius a is $\frac{2a}{\sqrt{3}}$.
45. Show that the volume of the largest cone that can be inscribed in a sphere of radius R is $\frac{8}{27}$ of the volume of the sphere.
46. Show that the volume of the largest cylinder which can be inscribed in a cone of height h and semi vertical angle α is $\frac{4}{27} \times \pi h^3 \tan^2 \alpha$.

Co-ordinate Geometrical Problems

47. Find the point on the curve $y^2 = 4x$ which is closest to the point $(2, 1)$.
48. A jet of an enemy is flying along the curve $y = x^2 + 2$. A soldier is placed at the point $(3, 2)$. Find the shortest distance between the soldier and the jet.
49. Find the shortest distance of the point $(0, c)$ from the parabola $y = x^2$, where $0 \leq c \leq 5$.
50. Find the point on the curve $4x^2 + a^2y^2 = 4a^2$, $4 < a^2 < 8$ that is farthest from the point $(0, -2)$.
51. What normal to the curve $y = x^2$ forms the shortest chord?
52. Find a point on the curve $x^2 + 2y^2 = 6$ whose distance from the line $x + y = 7$ is minimum.
53. Find the shortest distance between the curves $x^2 + y^2 = 2$ and $xy = 9$.

54. Find the shortest distance between the curves

$$y^2 = 4x \text{ and } x^2 + y^2 + 12x + 12y + 12 = 0$$

Miscellaneous Problems

55. Find the least value of
 $f(x) = ax + \frac{b}{x}$, $a, b, x > 0$
56. Find the least value of
 $f(x) = x^2 + \frac{1}{x^2 + 1}$
57. Find the least value of
 $f(x) = \frac{x^3 + x + 2}{x}$, $x > 0$
58. Find the least value of
 $f(x) = 2\cos x + \sec^2 x$, $x \in \left[0, \frac{\pi}{2}\right)$
59. Find the least value of
 $f(x) = 2\log_{10} x - \log x(0.01)$, $x > 1$
60. Find the min values of
 $f(x) = \frac{a^2}{\sin^2 x} + \frac{b^2}{\cos^2 x}$
61. Find the min values of
 $f(x) = 2^x + 3^x + 5^x + \frac{1}{2^x} + \frac{1}{3^x} + \frac{1}{5^x}$, $x > 0$
62. Find the Min values of
 $f(a) = a^{-5} + a^{-4} + 3a^{-3} + 1 + a^8 + a^{10}$, $a > 0$
63. Find the min values of
 $f(x) = x^{10} + x^7 + \frac{2}{x^3} + \frac{4}{x^2} + \frac{3}{x}$, $x > 0$
64. Find the min value of
 $f(a, b, c, d) = \frac{(a^2 + 1)(b^2 + 1)(c^2 + 1)(d^2 + 1)}{abcd}$
 where $a, b, c, d > 0$.
65. Find the max value of x^2y^3 , where $2x + 3y = 5$.
66. Find the max value of x^3y^2z , where $3x + 2y + z = 14$.
67. Find the max value of xyz , if
 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.
68. Find the max value of
 $y = \frac{x}{ax^2 + b}$, $a, b, x > 0$
69. Find the least value of
 $f(x) = x^2 + 1 + \frac{4}{x^2 + 3}$
70. Find the least value of
 $f(x, y, z) = \frac{(x^3 + 2)(y^3 + 2)(z^3 + 2)}{xyz}$
 where $x, y, z > 0$

Level I

1. (i) We have $f(x) = x^2 - 4x + 10$

$$= (x - 2)^2 + 6 \geq 6$$

Thus, the min value of $f(x)$ is 6 at $x = 2$.

(ii) $f(x) = -x^2 + 6x - 15$

We have $f(x) = -x^2 + 6x - 15$

$$= -(x^2 - 6x + 9) - 6$$

$$= -(x - 3)^2 - 6 \leq -6$$

Thus, the maximum value of $f(x)$ is -6 at $x = 3$.

(iii) We have

$$f(x) = (x - 3)^2 + \geq 5$$

Hence, the min value of $f(x)$ is 5.

(iv) We have $f(x) = -(x - 1)^2 + 2 \leq 2$

Hence, the maximum value of $f(x)$ is 2.

(v) We have $f(x) = |x + 2|$ clearly, $|x + 2| \geq 0$

$$\Rightarrow f(x) \geq 0$$

Thus, the min value of $f(x)$ is 0 at $x = -2$

(vi) We have $f(x) = -|x - 1| + 5$

Now, $|x - 1| \geq 0$

$$\Rightarrow -|x - 1| \leq 0$$

$$\Rightarrow -|x - 1| + 5 \leq 5$$

$$\Rightarrow f(x) \leq 5$$

Thus, the maximum value of $f(x)$ is 5 at $x = 1$

(vii) We have $f(x) = |x - 1| + |x - 3|$

$$= \begin{cases} -(x-1) - (x-3) & : x < 1 \\ (x-1) - (x-3) & : 1 \leq x < 3 \\ (x-1) + (x-3) & : x \geq 3 \end{cases}$$

$$= \begin{cases} -2x + 4 & : x < 1 \\ 2 & : 1 \leq x < 3 \\ 2x - 4 & : x \geq 3 \end{cases}$$

Clearly, the min value of $f(x)$ is 2 at x in $[1, 3]$

(viii) We have $f(x) = |x + 2| + |x - 3|$

$$= \begin{cases} -2x + 1 & : x < -2 \\ 5 & : -2 \leq x < 3 \\ 2x - 1 & : x \geq 3 \end{cases}$$

Clearly, the min value of $f(x)$ is 5 at x in $[-2, 3]$

(ix) We have $f(x) = x^3 - 1$

Since the range of $f(x)$ is R , so it has no extrema.

(x) We have $f(x) = 3 \sin x + 4 \cos x + 10$

$$\text{Max value of } f(x) = \sqrt{3^2 + 4^2} + 10 = 5 + 10 = 15$$

$$\text{Min value of } f(x) = -\sqrt{3^2 + 4^2} + 10 = -5 + 10 = 5$$

(xi) We have $f(x) = \sin x + \cos x$

$$\text{Max value of } f(x) = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\text{Min value of } f(x) = -\sqrt{1^2 + 1^2} = -\sqrt{2}$$

(xii) We have $f(x) = \sin x - \cos x$

$$\text{Max value of } f(x) = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\text{Min value of } f(x) = -\sqrt{1^2 + 1^2} = -\sqrt{2}$$

(xiii) We have $f(x) = \sin(2x) + 5$

$$\text{Max value of } f(x) = 1 + 5 = 6$$

$$\text{Min value of } f(x) = -1 + 5 = 4$$

(xiv) We have $f(x) = 3 - 2\sin x = 3 + 2(-\sin x)$

$$\text{Max value of } f(x) = 3 + 2 \cdot 1 = 5$$

$$\text{Min value of } f(x) = 3 + 2(-1) = 3 - 2 = 1$$

(xv) We have $f(x) = 2\cos x + 3$

$$\text{Max value of } f(x) = 2 \cdot 1 + 3 = 5$$

$$\text{Min value of } f(x) = 2 \cdot (-1) + 3 = -2 + 3 = 1$$

(xvi) We have $f(x) = 3 - 4\cos x = 3 + 4(-\cos x)$

$$\text{Max value of } f(x) = 3 + 4 \cdot 1 = 7$$

$$\text{Min value of } f(x) = 3 + 4 \cdot (-1) = 3 - 4 = -1$$

(xvii) We have $f(x) = 2\sin^2 x + 4$

$$\text{Max value of } f(x) = 2 \cdot 1 + 4 = 6$$

$$\text{Min value of } f(x) = 2 \cdot 0 + 4 = 4$$

(xviii) We have $f(x) = 5 - 3\sin^2 x = 5 + 3(-\sin^2 x)$

$$\text{Max value of } f(x) = 5 + 3 \cdot 0 = 5$$

$$\text{Min value of } f(x) = 5 + 3(-1) = 5 - 3 = 2.$$

(xix) We have $f(x) = 3\sin^2 x + 2\cos^2 x$

$$= 2(\sin^2 x + \cos^2 x) + \sin^2 x$$

$$= 2 + \sin^2 x$$

$$\text{Max value of } f(x) = 2 + 1 = 3$$

$$\text{Min value of } f(x) = 2 + 0 = 2$$

(xx) We have

$$f(x) = 4\sin^2 x + 5\cos^2 x + 6\sin x \cos x + 10$$

$$= 4(\sin^2 x + \cos^2 x) + \cos^2 x + 3\sin 2x + 10$$

$$= 4 + \cos^2 x + 3\sin 2x + 10$$

$$= \cos^2 x + 3\sin 2x + 14$$

$$= \frac{1}{2}(1 + \cos 2x) + 3\sin 2x + 14$$

$$= \frac{1}{2}\cos 2x + 3\sin 2x + \frac{29}{2}$$

$$\text{Max value of } f(x) = \frac{\sqrt{37}}{2} + \frac{29}{2}$$

$$\text{Min value of } f(x) = -\frac{\sqrt{37}}{2} + \frac{29}{2}$$

2. (i) We have $f(x) = \sin(\sin x)$

$$\text{As we know that } -1 \leq \sin x \leq 1$$

$$\Rightarrow \sin(-1) \leq \sin(\sin x) \leq \sin(1)$$

$$\Rightarrow -\sin(1) \leq \sin(\sin x) \leq \sin(1)$$

$$\Rightarrow -\sin(1) \leq f(x) \leq \sin(1)$$

Min value of $f(x)$ is $\sin(1)$ and the max value of $f(x)$ is $-\sin(1)$

(ii) We have $f(x) = \cos(\cos x)$

$$\text{As we know that } -1 \leq \cos x \leq 1$$

$$\Rightarrow \cos(1) \leq \cos(\cos x) \leq 1$$

Thus, the max value of $f(x)$ is 1

and the min value of $f(x)$ is $\cos 1$

(iii) We have $f(x) = \sin(\sin x) + \cos(\sin x)$

As we know that,

$$-\sqrt{2} \leq \sin x + \cos x \leq \sqrt{2}$$

$$\Rightarrow -\sqrt{2} \leq \sin(\sin x) + \cos(\sin x) \leq \sqrt{2}$$

Thus, the max value of $f(x)$ is $\sqrt{2}$

and the min value of $f(x)$ is $-\sqrt{2}$

(iv) We have $f(x) = \sin^2 x + \cos^4 x$

$$= \frac{1}{2}(2\cos^2 x) + \frac{1}{4}(2\sin^2 x)^2$$

$$= \frac{1}{2}(1 + \cos 2x) + \frac{1}{4}(1 - \cos 2x)^2$$

$$= \frac{1}{2}(1 + \cos 2x) + \frac{1}{4}(1 - 2\cos 2x + \cos^2 2x)$$

$$= \frac{1}{2} + \frac{1}{4}(1 + \cos^2 2x)$$

$$= \frac{3}{4} + \frac{1}{4}\cos^2 2x$$

$$\text{Max value of } f(x) = \frac{3}{4} + \frac{1}{4} \cdot 1 = \frac{3}{4} + \frac{1}{4} = 1$$

$$\text{Min value of } f(x) = \frac{3}{4} + \frac{1}{4}(-1) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

(v) We have $f(x) = \sin^4 x + \cos^2 x$

$$= \frac{1}{4}(2\sin^2 x)^2 + \frac{1}{2}(2\cos^2 x)$$

$$= \frac{1}{4}(1 - \cos 2x)^2 + \frac{1}{2}(1 + \cos 2x)$$

$$= \frac{1}{4}(1 - 2\cos 2x + \cos^2 2x) + \frac{1}{2}(1 + \cos 2x)$$

$$= \frac{1}{4}(1 + \cos^2 2x) + \frac{1}{2}$$

$$= \frac{3}{4} + \frac{1}{4}\cos^2 2x$$

$$\text{Max value of } f(x) = \frac{3}{4} + \frac{1}{4} \cdot 1 = \frac{3}{4} + \frac{1}{4} = 1$$

$$\text{Min value of } f(x) = \frac{3}{4} + \frac{1}{4}(-1) = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$$

$$\begin{aligned} \text{(vi) We have } f(x) &= \sin^4 x + \cos^4 x \\ &= (\sin^2 x)^2 + (\cos^2 x)^2 \\ &= (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x \\ &= 1 - 2\sin^2 x \cos^2 x \\ &= 1 - \frac{1}{2}(4\sin^2 x \cos^2 x) \\ &= 1 - \frac{1}{2}(2\sin x \cos x)^2 \\ &= 1 - \frac{\sin^2 2x}{2} \\ &= 1 + \frac{1}{2}(-\sin^2 2x) \end{aligned}$$

$$\text{Max value of } f(x) = 1 + \frac{1}{2} \cdot 0 = 1$$

$$\text{Min value of } f(x) = 1 + \frac{1}{2} \cdot (-1) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\begin{aligned} \text{(vii) We have } f(x) &= \sin^6 x + \cos^6 x \\ &= 1 - 3\sin^2 x \cos^2 x \\ &= 1 - \frac{3}{4}\sin^2 2x \\ &= 1 + \frac{3}{4}(-\sin^2 2x) \end{aligned}$$

$$\text{Max value of } f(x) = 1 + \frac{3}{4} \cdot 0 = 1$$

$$\text{Min value of } f(x) = 1 + \frac{3}{4} \cdot (-1) = 1 - \frac{3}{4} = \frac{1}{4}$$

$$\begin{aligned} \text{(viii) We have } f(x) &= \sin^2(\sin x) + \cos^2(\cos x) \\ &= \sin^2(\cos x) + \cos^2(\cos x) + \sin^2(\sin x) \\ &\quad - \sin^2(\cos x) \\ &= 1 + \sin^2(\sin x) - \sin^2(\cos x) \end{aligned}$$

$$\text{Max value of } f(x) = 1 + \sin^2(1)$$

$$\text{Min value of } f(x) = 1 - \sin^2(1)$$

$$\text{(ix) We have } f(x) = |\sin x| + |\cos x|$$

$$\text{Max value of } f(x) = 1$$

$$\text{Min value of } f(x) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}$$

$$\text{(x) We have } f(x) = \sin x + \operatorname{cosec} x \text{ for all } x \text{ in } \left(0, \frac{\pi}{2}\right)$$

As we know that, A.M $\geq$ G.M

$$\Rightarrow \left(\frac{\sin x + \operatorname{cosec} x}{2} \right) \geq \sqrt{\sin x \cdot \operatorname{cosec} x}$$

$$\Rightarrow \left(\frac{\sin x + \operatorname{cosec} x}{2} \right) \geq 1$$

$$\Rightarrow \sin x + \operatorname{cosec} x \geq 2$$

Hence, the **Maxima & Minima** of $f(x)$ is 2.

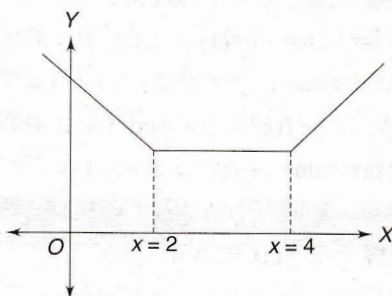
$$3. \text{ We have } f(x) = |x - a| + |x - b|$$

$$= \begin{cases} -(x-a) - (x-a) & : x < a \\ (x-a) - (x-b) & : a \leq x < b \\ (x-a) + (x-b) & : x \geq b \end{cases}$$

$$= \begin{cases} -2x + a + b & : x < a \\ b - a & : a \leq x < b \\ 2x - a - b & : x \geq b \end{cases}$$

Thus, the function has a min value $a - b$ at $x \in [a, b]$

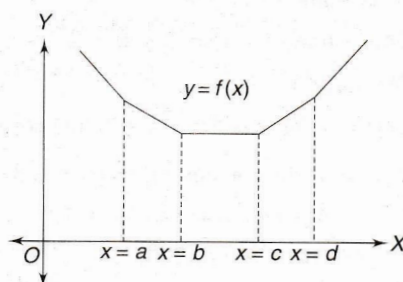
$$4. \text{ We have } f(x) = |x - 2| + |x - 4|$$



Thus, the function has a min value 2 at $x \in [2, 4]$

$$5. \text{ We have}$$

$$f(x) = \begin{cases} -4x + (a + b + c + d) & : x < a \\ 2x - a + (b + c + d) & : a \leq x < b \\ c + d - a - b & : b \leq x < c \\ 2x - (a + b + c) - d & : c \leq x < d \\ 4x - (a + b + c + d) & : x \geq d \end{cases}$$



From the graph, it is clear that, $f(x)$ has a **min** value

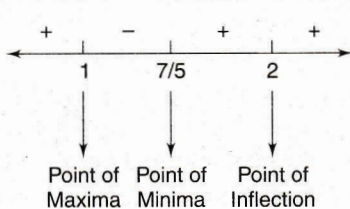
$$= c + d - a - b \text{ at } x \in [b, c]$$

$$6. \text{ We have } f(x) = (x - 1)^2(x - 2)^3$$

$$\Rightarrow f'(x) = 2(x - 1)(x - 2)^3 + 3(x - 1)^2(x - 2)^2$$

$$\Rightarrow f'(x) = (x - 1)(x - 2)^2(2(x - 2) + 3(x - 1))$$

$$\Rightarrow f'(x) = (x - 1)(x - 2)^2(5x - 7)$$



Thus, $f(x)$ has a local max value at $x = 1$ and local min value at $x = 7/5$.

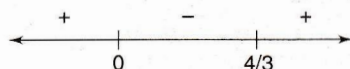
7. We have, $f(x) = (x - 2)x^{2/3}$

$$\Rightarrow f'(x) = x^{2/3} + (x - 2) \cdot \frac{2}{3} \cdot x^{-1/3}$$

$$\Rightarrow f'(x) = x^{2/3} + \frac{2(x - 2)}{3x^{1/3}}$$

$$\Rightarrow f'(x) = \frac{x + 2x - 4}{3\sqrt[3]{x}}$$

$$\Rightarrow f'(x) = \frac{3x - 4}{3\sqrt[3]{x}}$$



Thus, $f(x)$ has a local max value at $x = 0$ and local min value at $x = 4/3$.

8. Given $f(x) = x^3 - 6x^2 + 12x - 8$

$$\Rightarrow f'(x) = 3x^2 - 12x + 12$$

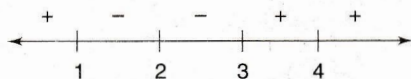
$$\Rightarrow f'(x) = 3(x^2 - 4x + 4) = 3(x - 2)^2$$

So, $f'(x)$ does not change its sign at the nbd of $x = 2$. Thus, $f(x)$ has no point of extremum.

Therefore, $x = 2$ is the point of inflection.

9. We have

$$f(x) = (x - 1)(x - 2)^2(x - 3)^3(x - 4)^4$$



Thus, $x = 1$ is the point of local max and $x = 3$ is the point of local min and the points $x = 2, 4$ are the point of inflections.

10. We have $f(x) = \sqrt{2x^2 - x + 2}$

$$\Rightarrow f'(x) = \frac{1}{2\sqrt{2x^2 - x + 2}} \times (4x - 1)$$

$$\Rightarrow f'(x) = \frac{(4x - 1)}{2\sqrt{2x^2 - x + 2}}$$

So, $f'(x)$ changes its sign at $x = 1/4$ from negative to positive, so $f(x)$ has a point of local min.

11. We have $f'(x) = 1 - \frac{2x}{(1 + x^2)}$

Maxima & Minima

$$\Rightarrow f'(x) = \frac{x^2 + 1 - 2x}{(1 + x^2)} = \frac{(x - 1)^2}{(x^2 + 1)}$$

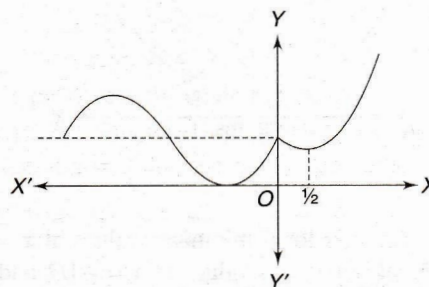
So, $f'(x)$ does not change its sign at the nbd of $x = 1$.

Thus, $f(x)$ has no point of local max or min.

12. We have $f(x) = \begin{cases} 1 + \sin x & : x < 0 \\ x^2 - x + 1 & : x \geq 0 \end{cases}$

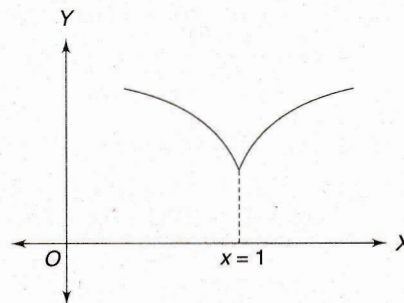
$$f'(x) = \begin{cases} \cos x & : x < 0 \\ 2x - 1 & : x \geq 0 \end{cases}$$

Thus, $f'(0^+) = 1 = f'(0^-)$.



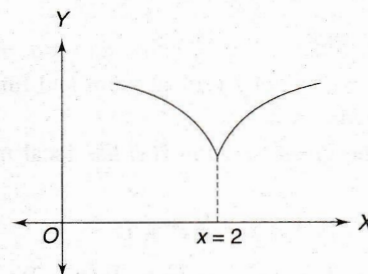
So, $f(x)$ has a local max value at $x = 0$

13. Given $f(x) = 1 + (x - 1)^{2/3}$



From the graph, it is clear that, $f'(x)$ changes its sign from negative to positive, so $x = 1$ is the point of minimum.

14. We have, $f(x) = 1 + (x - 2)^{4/5}$



So, $x = 2$ is the point of minimum.

Now, $f(0) = 0, f\left(\frac{\pi}{3}\right) = \frac{\pi}{3} + \frac{\sqrt{3}}{2},$

$$f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}, f(2\pi) = 2\pi$$

Therefore, the absolute max value = 2π and absolute min value = 0

23. Given $f(x) = \sin x + \frac{1}{2}\cos 2x$

$$\Rightarrow f'(x) = \cos x - \sin 2x$$

Now, $f'(x) = 0$ gives $\cos x - \sin 2x = 0$

$$\Rightarrow \cos x(1 - 2\sin x) = 0$$

$$\Rightarrow \cos x = 0, (1 - 2\sin x) = 0$$

$$\Rightarrow \cos x = 0, \sin x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{6}, \frac{\pi}{2}$$

Now, $f(0) = \frac{1}{2}, f\left(\frac{\pi}{6}\right) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4},$

$$f\left(\frac{\pi}{2}\right) = 1 - \frac{1}{2} = \frac{1}{2}$$

Therefore, the absolute max value = $3/4$ and absolute min = $1/2$.

24. Given $f(x) = \tan^{-1} x - \frac{1}{2}\ln x$

$$\Rightarrow f'(x) = \frac{1}{1+x^2} - \frac{1}{2x} = -\frac{x^2 - 2x + 1}{2x(x^2 + 1)}$$

Now, $f'(x) = 0$ gives $x = 1$

Thus, $f(1) = \frac{\pi}{4}, f(\sqrt{3}) = \frac{\pi}{3} - \frac{1}{4}\log 3,$

$$f\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} + \frac{1}{4}\log 3$$

Therefore, the max value = $\frac{\pi}{6} + \frac{1}{4}\log 3$ and min value = $\frac{\pi}{3} - \frac{1}{4}\log 3.$

25. Given $f(x) = \{(1 - x^2)(2x^2 + 1)\}^{1/2}$

$$\Rightarrow f'(x) = \frac{(1 - x^2) \cdot 4x + (2x^2 + 1) \cdot (2x)}{2\sqrt{(1 - x^2)(2x^2 + 1)}}$$

$$\Rightarrow f'(x) = \frac{2x(2x^2 + 1 + 2 - 2x^2)}{2\sqrt{(1 - x^2)(2x^2 + 1)}}$$

$$\Rightarrow f'(x) = \frac{6x}{2\sqrt{(1 - x^2)(2x^2 + 1)}}$$

Now, $f'(x) = 0$ gives $x = 0.$

Thus, $f(0) = 1, f(-1) = 0, f(1) = 0$

Therefore, the max value = 1 and min value = 0.

26. Given $f(x) = \frac{\sin 2x}{\sin\left(x + \frac{\pi}{4}\right)}$ **Maxima & Minima**

Max value occurs at $x = \frac{\pi}{4}$

So, Max value = 1

Min value occurs at $x = 0.$

Min value = 0

27. Given $f(x) = \cos x + \cos(\sqrt{2}x)$

The maximum value of $f(x)$ occurs when

$$\Rightarrow \cos x = 1 \text{ \& } \cos(x\sqrt{2}) = 1$$

$$\Rightarrow x = 2n\pi \text{ \& } x\sqrt{2} = 2m\pi, m, n \in I$$

$$\Rightarrow x = 2n\pi \text{ \& } x = \sqrt{2}m\pi, m, n \in I$$

$$\Rightarrow 2n\pi = \sqrt{2}m\pi$$

$$\Rightarrow m = \sqrt{2}n$$

It is possible only when $x = 0.$

Thus, the function will provide us the maximum value at $x = 0$ and the maximum value is 2.

28. We have $f(x) = \frac{1}{\sin x + 4} + \frac{1}{\cos x - 4}$

$$\Rightarrow f'(x) = \frac{\cos x}{(\sin x + 4)^2} - \frac{\sin x}{(\cos x - 4)^2}$$

Now $f'(x) = 0$ gives

$$\Rightarrow \frac{\cos x}{(\sin x + 4)^2} - \frac{\sin x}{(\cos x - 4)^2} = 0$$

$$\Rightarrow \frac{\cos x}{(\sin x + 4)^2} = \frac{\sin x}{(\cos x - 4)^2}$$

$$\Rightarrow \cos x(\cos x - 4)^2 = \sin x(\sin x + 4)^2$$

$$\Rightarrow \cos x(\cos^2 x - 8\cos x + 16)$$

$$= \sin x(\sin^2 x + 8\sin x + 16)$$

$$\Rightarrow (\cos^3 x - \sin^3 x) - 8(\cos^2 x - \sin^2 x)$$

$$+ 16(\cos x - \sin x) = 0$$

$$\Rightarrow (\cos x - \sin x) \times \{(1 - \sin x \cos x)$$

$$- 8(\cos x + \sin x) + 16\} = 0$$

$$\Rightarrow \tan x = \frac{\pi}{4} \text{ and put } \sin x + \cos x = t$$

Hence, the min value of the function

$$= \frac{1}{\frac{1}{\sqrt{2}} + 4} + \frac{1}{\frac{1}{\sqrt{2}} - 4}$$

$$= \frac{4 - \frac{1}{\sqrt{2}}}{16 - \frac{1}{2}} - \frac{4 + \frac{1}{\sqrt{2}}}{16 - \frac{1}{2}}$$

$$= \frac{2}{31} \left(4 - \frac{1}{\sqrt{2}} - 4 - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{2}{31} \times \frac{-2}{\sqrt{2}}$$

$$= -\frac{4}{31\sqrt{2}}$$

29. Given $f(x) = \frac{14}{x^4 - 8x^2 + 2}$

$$= \frac{14}{(x^2 - 4)^2 - 14}$$

Hence, the global max value is -1 at $x = \pm 2$ and global min value is 7 at $x = 0$.

30. Given $f(x) = (x-1)^2 \sqrt{x^2 - 2x + 3}$

$$\Rightarrow f'(x) = 2(x-1) \sqrt{x^2 - 2x + 3} + (x-1)^2 \times \frac{x-1}{\sqrt{x^2 - 2x + 3}}$$

$$\Rightarrow f'(x) = 2(x-1) \sqrt{x^2 - 2x + 3} + \frac{(x-1)^3}{\sqrt{x^2 - 2x + 3}}$$

$$\Rightarrow f'(x) = \frac{2(x-1)(x^2 - 2x + 3) + (x-1)^3}{\sqrt{x^2 - 2x + 3}}$$

$$f'(x) = 0 \text{ gives}$$

$$\Rightarrow \frac{2(x-1)(x^2 - 2x + 3) + (x-1)^3}{\sqrt{x^2 - 2x + 3}} = 0$$

$$\Rightarrow (x-1)(3x^2 - 6x + 7) = 0$$

$$\Rightarrow x = 1$$

Now, $f(1) = 0, f(0) = \sqrt{3}, f(3) = 4\sqrt{6}$

Thus, the global max value $= 4\sqrt{6}$ and global min is 0 .

31. Given $y = \frac{x^2 - 7x + 6}{x - 10}$

$$\Rightarrow \frac{dy}{dx} = \frac{(x-10)(2x-7) - (x^2 - 7x + 6) \cdot 1}{(x-10)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 - 20x + 64}{(x-10)^2} = \frac{(x-10)^2 - 36}{(x-10)^2}$$

$$\Rightarrow \frac{dy}{dx} = 1 - \frac{36}{(x-10)^2}$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{72}{(x-10)^2}$$

Now, $\frac{dy}{dx} = 0$ gives $x = 4, 16$

Maxima & Minima

At $x = 4, \frac{d^2y}{dx^2} = -\frac{1}{3} < 0$

Thus, y is max at $x = 4$ and the max value $= 1$

At $x = 16, \frac{d^2y}{dx^2} = \frac{1}{3} > 0$

Thus, y is min at $x = 16$ and the min value $= 25$.

32. Given $f(x) = 2 \sin x + \cos 2x$

$$\Rightarrow f'(x) = 2 \cos x - 2 \sin 2x$$

$$\Rightarrow f''(x) = -2 \sin x - 4 \cos 2x$$

Now $f'(x) = 0$ gives

$$\Rightarrow 2 \cos x - 2 \sin 2x = 0$$

$$\Rightarrow 2 \cos x - 4 \sin x \cos x = 0$$

$$\Rightarrow 2 \cos x (1 - 2 \sin x) = 0$$

$$\Rightarrow 2 \cos x = 0, (1 - 2 \sin x) = 0$$

$$\Rightarrow \cos x = 0, \sin x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}$$

At $x = \frac{\pi}{6}, f''(x) = -1 - \sqrt{3} < 0$

Thus, f is max at $x = \frac{\pi}{6}$ and the max value is

$$= 2 \cdot \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

At $x = \frac{\pi}{2}, f''(x) = -2 + 4 = 2 > 0$

Thus, f is min at $x = \frac{\pi}{2}$ and the min value is $= 2 - 1 = 1$.

Also, at $x = \frac{5\pi}{6}, f''(x) = -2 \cdot \frac{1}{2} - 4 \cdot \frac{1}{2} = -3$

Thus, f is max at $x = \frac{5\pi}{6}$ and the max value is $= 3/2$

Hence, the max value $= 3/2$ and min value $= 1$.

33. Given $f(x) = 3x^4 - 2x^3 - 3x^2 + 10$

$$\Rightarrow f'(x) = 12x^3 - 6x^2 - 6x$$

$$\Rightarrow f'(x) = 6(2x^3 - x^2 - x)$$

$$\Rightarrow f''(x) = 6(6x^2 - 2x - 1)$$

Now, $f'(x) = 0$ gives $x(2x^2 - x - 1) = 0$

$$\Rightarrow x(x-1)(2x+1) = 0$$

$$\Rightarrow x = 0, 1, -1/2$$

At $x = 0, f''(x) = -6 < 0$,

Thus f is max and the max value $= f(0) = 10$

At $x = 1$, $f''(x) = 18 > 0$, so f is min and the min value $= f(1) = 8$.

At $x = -1/2$, $f''(x) = 6\left(\frac{6}{4} + 2 \cdot \frac{1}{2} - 1\right) = 9 > 0$

so f is min and the min value

$$= f\left(-\frac{1}{2}\right) = \frac{3}{16} + \frac{2}{8} - \frac{3}{4} + 10$$

$$= 10 - \frac{5}{16} = \frac{155}{16} = 9\frac{11}{16}$$

Hence, the max value $= 10$ and the min value $= 9$.

34. Given $f(x) = \ln(x^4 - 2x^2 + 3)$

Let $g(x) = (x^4 - 2x^2 + 3)$

So g is max or min, then f is max or min.

Now, $g'(x) = 4x^3 - 4x$

$$\Rightarrow g''(x) = 12x^2 - 4$$

$$\Rightarrow g''(x) = 4(3x^2 - 1)$$

Now, $g'(x) = 0$ gives $x^3 - x = 0$

$$\Rightarrow x(x+1)(x-1) = 0$$

$$\Rightarrow x = 0, -1, 1.$$

At $x = 0$, $g''(x) = -4 < 0$, so g is max

Thus f is max at $x = 0$ and the max value $= \ln 3$

At $x = -1$, $g''(x) = 8 > 0$, so g is min

Thus f is min at $x = -1$ and the min value $= \ln 2$

At $x = 1$, $g''(x) = 8 > 0$, so g is min.

Thus f is min at $x = 1$ and the min value $= \ln 2$.

Hence the min value $= \ln 2$ and max value $= \ln 3$.

35. Given $f(x) = \sqrt{3x^2 - 2x^3}$

Let $g(x) = 3x^2 - 2x^3$

So g is max or min, then f is max or min.

$$\Rightarrow g'(x) = 6x - 6x^2$$

$$\Rightarrow g''(x) = 6 - 12x$$

Now, $g'(x) = 0$ gives $x - x^2 = 0$

$$\Rightarrow x = 0, 1$$

At $x = 0$, $g''(x) = 6 > 0$, g is min

Thus, f is min and the min value $= 0$

At $x = 1$, $g''(x) = 6 - 12 = -6 < 0$, g is max

Thus f is max and the max value $= 1$.

36. Given $f(x) = \frac{10}{3x^4 + 4x^3 - 12x^2 + 11}$

Let $g(x) = 3x^4 + 4x^3 - 12x^2 + 11$

So g is max or min, then f is max or min

Now, $g'(x) = 12x^3 + 12x^2 - 24x$

$$\Rightarrow g'(x) = 12(x^3 + x^2 - 2x)$$

$$\Rightarrow g''(x) = 12(3x^2 + 2x - 2)$$

Now, $g'(x) = 0$ gives $x(x^2 + x - 2) = 0$

$$\Rightarrow x(x+2)(x-1) = 0$$

$$\Rightarrow x = -2, 0, 1.$$

At $x = -2$, $g''(x) = 72 > 0$, so g is min

Thus, f is max and the max value $= \frac{1}{21}$

At $x = 0$, $g''(x) = -24 < 0$, g is max

Thus f is min and the min value is $= \frac{1}{11}$

At $x = 1$, $g''(x) = 36 > 0$, so g is min

Thus, f is max and the max value $= \frac{1}{6}$

Hence, the max value $= \frac{1}{6}$ and min value $= \frac{1}{11}$.

37. Given $f(x) = x^5 + 5x^4 + 5x^3 - 1$

$$\Rightarrow f'(x) = 5x^4 + 20x^3 + 15x^2$$

$$\Rightarrow f'(x) = 5x^2(x+1)(x+3)$$

$$\Rightarrow f''(x) = 5(4x^3 + 12x^2 + 6x)$$

$$\Rightarrow f'''(x) = 5(12x^2 + 24x + 6)$$

Now, $f'(x) = 0$ gives $x = -3, -1, 0$

At $x = -3$, $f''(x) = -90 < 0$

So, f is max and the max value $= 26$

At $x = -1$, $f''(x) = 10 > 0$

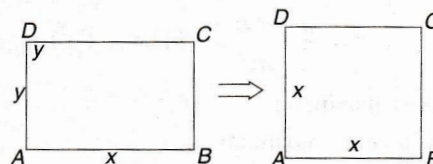
So f is min and the min value $= -2$

At $x = 0$, $f''(x) = 0$

Now, $f'''(0) = 30$.

So f has neither max nor min at $x = 0$.

38.



Let x and y be the sides of a rectangle.

Given $P = 2(x + y)$

$$\Rightarrow y = \left(\frac{P}{2} - x\right)$$

Now, $A = xy = x\left(\frac{p}{2} - x\right)$

$$\frac{dA}{dx} = \left(\frac{p}{2} - x\right) \cdot x + x(-1) = \left(\frac{p}{2} - 2x\right)$$

$$\frac{d^2A}{dx^2} = -2 < 0$$

So area is maximum.

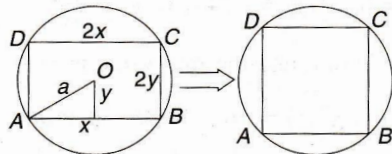
For max or min, $\frac{dA}{dx} = 0$ gives $x = \frac{p}{4}$

when $x = \frac{p}{4}$, then $y = \frac{p}{2} - \frac{p}{4} = \frac{p}{4}$

$\Rightarrow x = y$

$\Rightarrow$ Rectangles becomes square.

39.



Let the lengths of the sides be $2x$ and $2y$ respectively

and a be the radius of the circle.

Now, $x^2 + y^2 = a^2$

$$\Rightarrow y = \sqrt{a^2 - x^2}$$

We have, $A = 4xy = 4x\sqrt{a^2 - x^2}$

$$\Rightarrow A^2 = 16x^2(a^2 - x^2)$$

$$\Rightarrow B = 16(a^2x^2 - x^4)$$

$$\Rightarrow \frac{dB}{dx} = 16(2a^2x - x^3)$$

$$\Rightarrow \frac{d^2B}{dx^2} = 16(2a^2 - 12x^2)$$

For max or min, $\frac{dB}{dx} = 0$ gives

$$\Rightarrow 2a^2x - 4x^3 = 0$$

$$\Rightarrow a^2 - 2x^2 = 0$$

$$\Rightarrow x = \frac{a}{\sqrt{2}}$$

$$\text{At } x = \frac{a}{\sqrt{2}}, \frac{d^2B}{dx^2} = 16(2a^2 - 6a^2) = -64a^2 < 0$$

So, B is maximum

Thus, area is maximum.

when $x = \frac{a}{\sqrt{2}}$, then $y = \sqrt{a^2 - \frac{a^2}{2}} = \frac{a}{\sqrt{2}}$

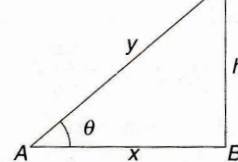
$$\Rightarrow x = y$$

$$\Rightarrow 2x = 2y$$

$\Rightarrow$ Rectangles become squares

40. Let base = x and hypotenuse = y resp and $x + y = k$

Maxima & Minima



Here, $x^2 + h^2 = y^2$

$$\Rightarrow h^2 = y^2 - x^2$$

$$\Rightarrow h = \sqrt{y^2 - x^2}$$

Area of the triangle

$$= A = \frac{1}{2} \times x \times h = \frac{1}{2} \sqrt{y^2 - x^2}$$

$$\Rightarrow A = \frac{x}{2} \sqrt{(k-x)^2 - x^2}$$

$$\Rightarrow A = \frac{x}{2} \sqrt{k^2 - 2kx}$$

$$\Rightarrow B = A^2 = \frac{x^2}{4} (k^2 - 2kx)$$

$$\Rightarrow B = \frac{1}{4} (k^2x^2 - 2kx^3)$$

$$\Rightarrow \frac{dB}{dx} = \frac{1}{4} (2k^2x - 6kx^2)$$

$$\Rightarrow \frac{d^2B}{dx^2} = \frac{1}{4} (2k^2 - 12kx)$$

For max or min, $\frac{dB}{dx} = 0$ gives $x = \frac{k}{3}$

$$\text{At } x = \frac{k}{3}, \frac{d^2B}{dx^2} = \frac{1}{4} (2k^2 - 4k^2) = -\frac{k^2}{2} < 0$$

So, B is maximum

Thus, area is maximum

when $x = \frac{k}{3}$, then $y = k - x = \frac{2k}{3}$

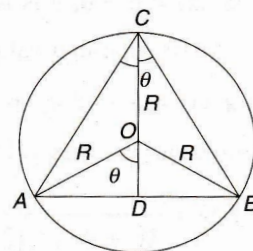
Let θ be the angle between them

$$\Rightarrow \cos \theta = \frac{x}{y} = \frac{k/3}{2k/3} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

Hence, the result.

41.



Here, $\cos \theta = \frac{OD}{R}$ and $\sin \theta = \frac{AD}{R}$

$\Rightarrow OD = R \cos \theta, AD = R \sin \theta$

We have, $A = \frac{1}{2} \cdot AB \cdot CD$

$\Rightarrow A = \frac{1}{2} \cdot (2R \sin \theta) \cdot (R + R \cos \theta)$

$\Rightarrow A = R^2 \cdot (\sin \theta + \sin \theta \cos \theta)$

$\Rightarrow \frac{dA}{d\theta} = R^2 \cdot (\cos \theta + \cos 2\theta)$

$\Rightarrow \frac{d^2A}{d\theta^2} = R^2 \cdot (-\sin \theta - 2 \sin 2\theta)$

For max or min, $\frac{dA}{d\theta} = 0$ gives

$\Rightarrow \cos \theta + \cos 2\theta = 0$

$\Rightarrow 2 \cos^2 \theta - 1 + \cos \theta = 0$

$\Rightarrow 2 \cos^2 \theta + \cos \theta - 1 = 0$

$\Rightarrow 2 \cos^2 \theta + 2 \cos \theta - \cos \theta - 1 = 0$

$\Rightarrow (2 \cos \theta - 1)(1 + \cos \theta) = 0$

$\Rightarrow \cos \theta = \frac{1}{2}, -1$

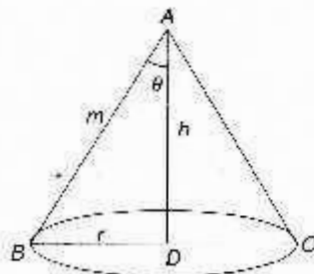
$\Rightarrow \theta = \frac{\pi}{3}$

At $\theta = \frac{\pi}{3}, \frac{d^2A}{d\theta^2} < 0$

So area is maximum.

Thus, the triangle is equilateral, where $\theta = \frac{\pi}{3}$.

42.



Let m = slant height, radius = r of the cone

Also, $\sin \theta = \frac{r}{m}, \cos \theta = \frac{h}{m}$

$\Rightarrow r = m \sin \theta, h = m \cos \theta$

Volume of the cone = V

$= \frac{1}{3} \pi r^2 h$

$= \frac{1}{3} \pi (m \sin \theta)^2 (m \cos \theta)$

$= \frac{m^3 \pi}{3} (\sin^2 \theta \cos \theta)$

Maxima & Minima

$= \frac{m^3 \pi}{3} (\cos \theta - \cos^3 \theta)$

$\Rightarrow \frac{dV}{d\theta} = \frac{m^3 \pi}{3} (-\sin \theta + 3 \cos^2 \theta \cdot \sin \theta)$

$\Rightarrow \frac{dV}{d\theta} = \frac{m^3 \pi}{3} (2 \sin \theta - \sin^3 \theta)$

$\Rightarrow \frac{d^2V}{d\theta^2} = \frac{m^3 \pi}{3} (2 \cos \theta - 9 \sin^2 \theta \cos \theta)$

For max or min, $\frac{dV}{d\theta} = 0$ gives $2 \sin \theta - 3 \sin^3 \theta = 0$

$\Rightarrow \sin \theta = \sqrt{\frac{2}{3}}$

$\Rightarrow \tan \theta = \sqrt{2}$

$\Rightarrow \theta = \tan^{-1}(\sqrt{2})$

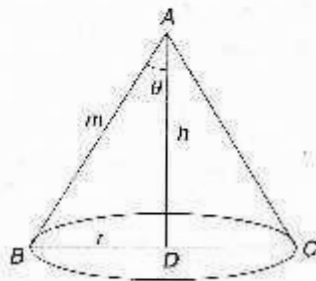
At $\sin \theta = \sqrt{\frac{2}{3}}, \frac{d^2V}{d\theta^2} < 0$

So, the volume is maximum.

Hence, the semi vertical angle of a right circular cone

$= \tan^{-1}(\sqrt{2})$.

43.



Let r be the radius, m be the slant height and h be the height of the right circular cone.

Given surface area - constant

$\Rightarrow \pi r^2 + \pi r m = k$

$m = \frac{k - \pi r^2}{\pi r}$

Volume of the cone

$= V = \frac{1}{3} \pi r^2 h$

$= \frac{1}{3} \pi r^2 \sqrt{m^2 - r^2}$

$\Rightarrow V^2 = \frac{\pi^2}{9} \times r^4 (m^2 - r^2)$

$\Rightarrow \dot{V} = \frac{\pi^2}{9} \times (m^2 r^4 - r^6)$

$$\Rightarrow z = \frac{\pi^2}{9} \times \left(r^4 \left(\frac{k - \pi r^2}{\pi r} \right)^2 - r^6 \right)$$

$$\Rightarrow z = \frac{\pi^2}{9} \times \left(r^4 \left(\frac{k^2 - 2k\pi r^2 + \pi^2 r^4}{\pi^2 r^2} \right) - r^6 \right)$$

$$\Rightarrow z = \frac{\pi^2}{9} \times \frac{1}{\pi^2 r^2} \times (r^4(k^2 - 2k\pi r^2 + \pi^2 r^4) - \pi^2 r^8)$$

$$\Rightarrow z = \frac{1}{9} \times (r^2(k^2 - 2k\pi r^2))$$

$$\Rightarrow z = \frac{1}{9} \times (k^2 r^2 - 2k\pi r^4)$$

$$\Rightarrow \frac{dz}{dr} = \frac{1}{9} \times (2k^2 r - 8k\pi r^3)$$

$$\Rightarrow \frac{d^2z}{dr^2} = \frac{1}{9} \times (2k^2 - 24k\pi r^2)$$

For max or min, $\frac{dz}{dr} = 0$ gives

$$2k^2 r - 8k\pi r^3 = 0$$

$$\Rightarrow k - 4\pi r^2 = 0$$

$$\Rightarrow r = \sqrt{\frac{k}{4\pi}}$$

At $r = \sqrt{\frac{k}{4\pi}}, \frac{d^2z}{dr^2} < 0$

So, z is maximum

Thus, volume is maximum.

when $r = \sqrt{\frac{k}{4\pi}}$, then $m = \frac{3}{2} \sqrt{\frac{k}{\pi}}$

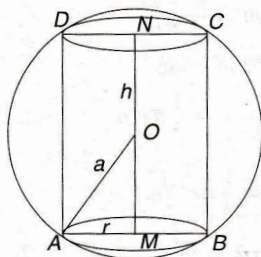
Let θ be the semi vertical angle of the cone.

Then $\sin \theta = \frac{r}{m} = \frac{\frac{1}{2} \sqrt{\frac{k}{\pi}}}{\frac{3}{2} \sqrt{\frac{k}{\pi}}} = \frac{1}{3}$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{1}{3}\right)$$

Hence, the result.

44.



Here, a = radius of the sphere

r = radius and h = height of the cylinder resp.

Now, $r^2 + \frac{h^2}{4} = a^2$

Maxima & Minima

$$\Rightarrow h = 2\sqrt{a^2 - r^2}$$

Volume of the cylinder

$$= V = \pi r^2 h = \pi r^2 2\sqrt{a^2 - r^2}$$

$$\Rightarrow V = 2\pi r^2 \sqrt{a^2 - r^2}$$

$$\Rightarrow z = V^2 = 4\pi^2 r^4 (a^2 - r^2)$$

$$\Rightarrow z = 4\pi^2 (a^2 r^4 - r^6)$$

$$\Rightarrow \frac{dz}{dr} = 4\pi^2 (4a^2 r^3 - 6r^5)$$

$$\Rightarrow \frac{d^2z}{dr^2} = 4\pi^2 (12a^2 r^2 - 30r^4)$$

For max or min, $\frac{dz}{dr} = 0$ gives

$$\Rightarrow 4a^2 r^3 - 6r^5 = 0$$

$$\Rightarrow 2a^2 - 3r^2 = 0$$

$$\Rightarrow r = \sqrt{\frac{2}{3}} a$$

When $r = \sqrt{\frac{2}{3}} a$, then

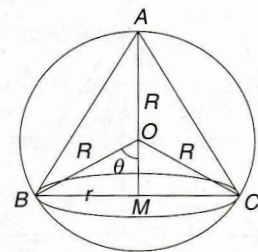
$$\Rightarrow h = 2\sqrt{a^2 - \frac{2}{3}a^2} = \frac{2a}{\sqrt{3}}$$

At $r = \sqrt{\frac{2}{3}} a, \frac{d^2z}{dr^2} < 0$

So z is maximum

Thus, Volume is maximum.

45.



Let R be the radius of the sphere and r be the radius of the cone.

$$\sin \theta = \frac{r}{R} \text{ and } \cos \theta = \frac{OM}{R}$$

Let V_1 be the volume of the sphere and V_2 be the volume of the cone

To prove, $V_2 = \frac{8}{27} \times V_1$

Now, $V_1 = \frac{4}{3} \pi R^3$

$$\begin{aligned}
\text{and } V_2 &= \pi r^2 h \\
\Rightarrow V_2 &= \pi r^2 (OA + OM) \\
\Rightarrow V_2 &= \pi R^2 \sin^2 \theta \cdot (R + R \cos \theta) \\
\Rightarrow V_2 &= \pi R^3 \cdot (\sin^2 \theta + \sin^2 \theta \cos \theta) \\
\Rightarrow \frac{dV_2}{d\theta} &= \pi R^3 (\sin 2\theta + \sin 2\theta \cos \theta - \sin^3 \theta) \\
\Rightarrow \frac{d^2 V_2}{d\theta^2} &= \pi R^3 [2 \cos 2\theta + 2 \cos 2\theta \cos \theta \\
&\quad - \sin 2\theta \sin \theta - 3 \sin^2 \theta \cos \theta]
\end{aligned}$$

For max or min, $\frac{dV_2}{d\theta} = 0$ gives

$$\begin{aligned}
\sin 2\theta + \sin 2\theta \cos \theta - \sin^3 \theta &= 0 \\
\Rightarrow 2 \cos \theta + 2 \cos^2 \theta - \sin^2 \theta &= 0 \\
\Rightarrow 3 \cos^2 \theta + 2 \cos \theta - 1 &= 0 \\
\Rightarrow (3 \cos \theta - 1)(\cos \theta + 1) &= 0 \\
\Rightarrow \cos \theta &= \frac{1}{3}
\end{aligned}$$

Thus, $\sin \theta = \frac{2\sqrt{2}}{3}$

At $\cos \theta = \frac{1}{3}$ and $\sin \theta = \frac{2\sqrt{2}}{3}$,

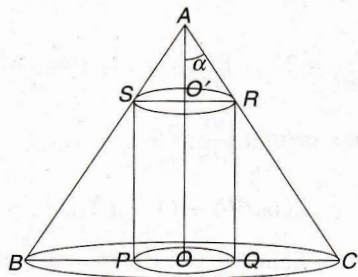
$$\Rightarrow \frac{d^2 V_2}{d\theta^2} < 0$$

So, the volume of the cone is maximum.

$$\begin{aligned}
\text{Thus, } V_2 &= \pi r^2 h = \pi (R \sin \theta)^2 (R + R \cos \theta) \\
&= \pi R^3 \cdot \sin^2 \theta \cdot (1 + \cos \theta) \\
&= \pi R^3 \left(\frac{2\sqrt{2}}{3} \right)^2 \cdot \frac{4}{3} \\
&= \pi R^3 \cdot \frac{8}{9} \cdot \frac{4}{3} \\
&= \frac{8}{27} \times \left(\frac{4}{3} \pi R^3 \right) \\
&= \frac{8}{27} \times V_1
\end{aligned}$$

Hence, the result.

46.



Let r be the radius of the cylinder

Now, $O'A = r \cot \alpha$ Maxima & Minima

Height of the cylinder $= OO' = h - r \cot \alpha$

Volume of the cylinder

$$\begin{aligned}
&= V = \pi r^2 (h - r \cot \alpha) \\
\Rightarrow V &= \pi (r^2 h - r^3 \cot \alpha) \\
\Rightarrow \frac{dV}{dr} &= \pi (2rh - 3r^2 \cot \alpha) \\
\Rightarrow \frac{d^2 V}{dr^2} &= \pi (2h - 6r \cot \alpha)
\end{aligned}$$

For max or min, $\frac{dV}{dr} = 0$ gives

$$\begin{aligned}
2rh - 3r^2 \cot \alpha &= 0 \\
\Rightarrow r &= \frac{2h}{3 \cot \alpha} \\
\Rightarrow r &= \frac{2h}{3 \tan \alpha}
\end{aligned}$$

At $r = \frac{2h}{3} \tan \alpha$, $\frac{d^2 V}{dh^2} = -2\pi h < 0$

So, volume is maximum

when $r = \frac{2h}{3} \tan \alpha$, then

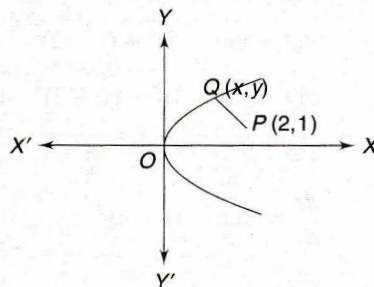
$$\text{height} = h - r \cot \alpha = h - \frac{2}{3} h = \frac{1}{3} h$$

Thus, Volume of the cylinder

$$\begin{aligned}
&= \pi \left(\frac{2h}{3} \tan \alpha \right)^2 \cdot \frac{h}{3} \\
&= \left(\frac{4}{27} \right) \times (\pi h^3 \tan^2 \alpha)
\end{aligned}$$

Hence, the result.

47.



Let $P = (2, 1)$ and $Q = (x, y)$

$$PQ = \sqrt{(x-2)^2 + (y-1)^2}$$

$$\Rightarrow PQ^2 = (x-2)^2 + (y-1)^2$$

$$\Rightarrow z = PQ^2 = \left(\frac{y^2}{4} - 2\right)^2 + (y-1)^2$$

$$\Rightarrow \frac{dz}{dy} = 2\left(\frac{y^2}{4} - 2\right)\left(\frac{2y}{4}\right) + 2(y-1)$$

$$\Rightarrow \frac{dz}{dy} = \left(\frac{y^3}{4} - 2y\right) + 2(y-1)$$

$$\Rightarrow \frac{dz}{dy} = \left(\frac{y^3}{4} - 2\right)$$

$$\Rightarrow \frac{d^2z}{dy^2} = \frac{3y^2}{4}$$

For max or min, $\frac{dz}{dy} = 0$

$$\Rightarrow \left(\frac{y^3}{4} - 2\right) = 0$$

$$\Rightarrow y = 2$$

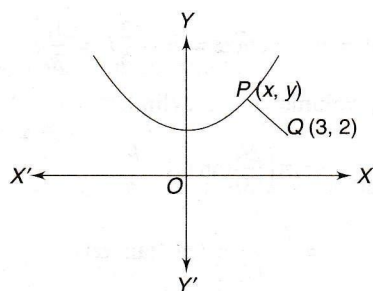
$$\text{At } y = 2, \frac{d^2z}{dy^2} = 3 > 0$$

So, z is minimum

Thus, distance is minimum when $y = 2$, then $x = 1$

Hence, the point $Q = (1, 2)$.

48. Given curve is $y = x^2 + 2$



Let the closest point of Q be $P(x, y)$

$$\text{Now, } PQ = \sqrt{(x-3)^2 + (y-2)^2}$$

$$\Rightarrow PQ^2 = (x-3)^2 + (y-2)^2$$

$$\Rightarrow z = PQ^2 = (x-3)^2 + x^4$$

$$\Rightarrow \frac{dz}{dx} = 2(x-3) + 4x^3$$

$$\Rightarrow \frac{d^2z}{dx^2} = 2 + 12x^2$$

For max or min, $\frac{dz}{dx} = 0$ gives

$$\Rightarrow 2(x-3) + 4x^3 = 0$$

$$\Rightarrow (x-3) + 2x^3 = 0$$

$$\Rightarrow 2x^3 + x - 3 = 0$$

$x = \text{Maxima \& Minima}$

$$\text{At } x = 1, \frac{d^2z}{dx^2} = 14 > 0$$

So, z is minimum Thus, the distance is minimum.

Hence, the minimum distance

$$= \sqrt{4+1} = \sqrt{5}$$

49. Given curve is $y = x^2$

Any point on the given parabola is (x, x^2)

Distance between (x, x^2) and $(0, c)$ is

$$z = \sqrt{(x^2)^2 + (x^2 - c)^2}$$

$$\Rightarrow z^2 = (x^2)^2 + (x^2 - c)^2 = x^4 - (2c-1)x^2 + c^2$$

$$= \left(x^2 - \left(\frac{2c-1}{2}\right)\right)^2 + \left(c^2 - \left(\frac{2c-1}{2}\right)^2\right)$$

$$= \left(x^2 - \left(\frac{2c-1}{2}\right)\right)^2 + \left(c^2 - \left(\frac{4c^2 - 4c + 1}{4}\right)\right)$$

$$= \left(x^2 - \left(\frac{2c-1}{2}\right)\right)^2 + \left(c - \frac{1}{4}\right)$$

Which is minimum when $x^2 = \left(\frac{2c-1}{2}\right)$ and the minimum value is

$$= z_{\min} = \sqrt{\left(c - \frac{1}{4}\right)}$$

50. Given curve is $4x^2 + a^2y^2 = 4a^2$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{4} = 1 \quad \dots(i)$$

Any point of (i) is $P(a \cos \theta, 2 \sin \theta)$ and given point is $(0, -2)$

$$\text{Let } z = PQ^2 = (a \cos \theta)^2 + (2 \sin \theta + 2)^2$$

$$\Rightarrow \frac{dz}{d\theta} = 2(a \cos \theta)(-a \sin \theta) + 2(2 \sin \theta + 2)(2 \cos \theta)$$

$$\Rightarrow \frac{dz}{d\theta} = 2 \cos \theta (4(1 + \sin \theta) - a^2 \sin \theta)$$

$$\Rightarrow \frac{dz}{d\theta} = 2 \cos \theta (4 + (4 - a^2) \sin \theta)$$

$$\Rightarrow \frac{d^2z}{d\theta^2} = 2 \cos^2 \theta (4 - a^2) - 2(4 - a^2) \sin \theta \sin \theta$$

For max or min, $\frac{dz}{d\theta} = 0$

$$\Rightarrow 2 \cos \theta (4 + (4 - a^2) \sin \theta) = 0$$

$$\Rightarrow 2 \cos \theta = 0, (4 + (4 - a^2) \sin \theta) = 0$$

$$\Rightarrow \cos \theta = 0, \sin \theta = \frac{4}{4-a^2}$$

$$\Rightarrow \theta = \frac{\pi}{2}, \sin \theta = \frac{4}{4-a^2}$$

$$\Rightarrow \theta = \frac{\pi}{2}, \sin \theta = \frac{4}{4-a^2} > 1$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

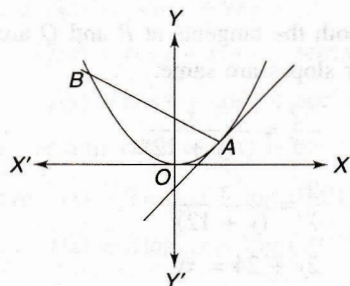
$$\Rightarrow \left(\frac{d^2z}{d\theta} \right)_{\theta=\frac{\pi}{2}} = 2(a^2 - 8) < 0$$

So z is maximum

Thus, distance is maximum.

Hence, the co-ordinates of the point P be $(0, 2)$

51. Given curve is $y = x^2$



Let $A = (t, t^2)$

Equation of the tangent at $A = (t, t^2)$ is

$$y - t^2 = \left(-\frac{1}{2t} \right) (x - t)$$

Let the normal intersects the curve again at B

where $B = (t_1, t_1^2)$

$$\text{Thus, } (t_1^2 - t^2) = \left(-\frac{1}{2t} \right) (t_1 - t)$$

$$\Rightarrow (t_1 + t) = \left(-\frac{1}{2t} \right)$$

$$\Rightarrow t_1 = -\left(t + \frac{1}{2t} \right)$$

$$\text{Now, } AB = \sqrt{(t - t_1)^2 + (t^2 - t_1^2)^2}$$

$$\Rightarrow AB^2 = (t - t_1)^2 + (t^2 - t_1^2)^2$$

$$\Rightarrow AB^2 = (t - t_1)^2 (1 + (t + t_1)^2)$$

$$\Rightarrow L = \left(2t + \frac{1}{2t} \right)^2 \left(1 + \frac{1}{4t^2} \right)$$

$$\Rightarrow L = 4t^2 \left(1 + \frac{1}{4t^2} \right)^3$$

$$\frac{dL}{dt} = 8t \left(1 + \frac{1}{4t^2} \right)^3 - 12t^2 \left(1 + \frac{1}{4t^2} \right)^2 \left(\frac{2}{4t^3} \right)$$

Maxima & Minima

$$\frac{dL}{dt} = 4 \left(1 + \frac{1}{4t^2} \right)^2 \left(2t - \frac{1}{t} \right)$$

$$\text{For max/min, } \frac{dL}{dt} = 0$$

$$\Rightarrow 4 \left(1 + \frac{1}{4t^2} \right)^2 \left(2t - \frac{1}{t} \right) = 0$$

$$\Rightarrow t^2 = \frac{1}{2}$$

$$\Rightarrow t = \pm \frac{1}{\sqrt{2}}$$

$$\left(\frac{d^2L}{dt^2} \right)_{t=\pm\frac{1}{\sqrt{2}}} = 16 \times \frac{9}{4} = 36 > 0$$

So L is minimum.

When $t = \frac{1}{\sqrt{2}}$ then

$$A = \left(\frac{1}{\sqrt{2}}, \frac{1}{2} \right) \text{ \& } B = (-\sqrt{2}, 2) \text{ and } AB = \frac{27}{4}$$

Thus, the equation of AB is

$$\frac{y-2}{\frac{1}{2}-2} = \frac{x+2}{\frac{1}{\sqrt{2}}+\sqrt{2}}$$

$$\Rightarrow \frac{y-2}{\frac{1}{\sqrt{2}}-\sqrt{2}} = \frac{x+2}{1}$$

$$\Rightarrow \sqrt{2}x + 2y - 2 = 0$$

When $t = -\frac{1}{\sqrt{2}}$, then

$$A = \left(-\frac{1}{\sqrt{2}}, \frac{1}{2} \right) \text{ \& } B = (\sqrt{2}, 2) \text{ and } AB = \frac{27}{4}$$

Thus, the equation of AB is

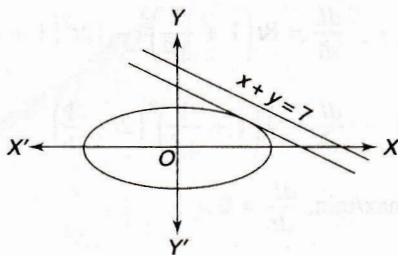
$$\frac{y-2}{\frac{1}{2}-2} = \frac{x-\sqrt{2}}{-\left(\frac{1}{\sqrt{2}}+\sqrt{2} \right)}$$

$$\Rightarrow \frac{y-2}{\frac{1}{\sqrt{2}}-\sqrt{2}} = \frac{x-\sqrt{2}}{-1}$$

$$\Rightarrow \sqrt{2}x - 2y + 2 = 0$$

52. Given curve is $x^2 + 2y^2 = 6$

$$\Rightarrow \frac{x^2}{6} + \frac{y^2}{3} = 1$$



Since the tangent is parallel to the line, so their slopes are same

$$\frac{dy}{dx} = -1$$

$$\Rightarrow \frac{-x}{2y} = -1$$

$$\Rightarrow x = 2y$$

when $x = 2y$, then $4y^2 + 2y^2 = 6$

$$\Rightarrow 6y^2 = 6$$

$$\Rightarrow y^2 = 1$$

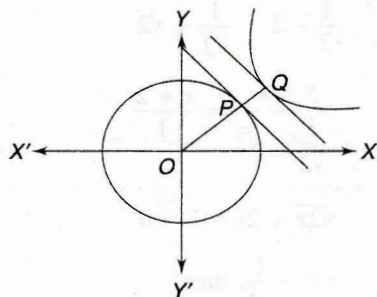
$$\Rightarrow y = \pm 1$$

when $x = \pm 1$, then $x = \pm 2$

So, the point on the curve is

$$(2, 1) \text{ or } (-2, -1).$$

53.



Given curves are $x^2 + y^2 = 2$ and $xy = 9$

Since both the tangents at P and Q are parallel, so their slopes are same

$$-\frac{x}{y} = -\frac{9}{x^2}$$

$$\Rightarrow x^3 = 9y$$

$$\Rightarrow x^3 = 9 \cdot \frac{9}{x} = \frac{81}{x}$$

$$\Rightarrow x^4 = 81$$

$$\Rightarrow x = 3$$

when $x = 3$, then $y = 3$

Thus, the point Q = (3, 3)

Therefore, the shortest distance

$$= PQ$$

$$= OQ - OP$$

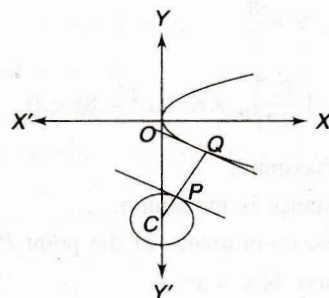
$$= \sqrt{9 + 9} - \sqrt{2}$$

$$= 3\sqrt{2} - \sqrt{2}$$

$$= 2\sqrt{2}$$

54. Given curves are

$$y^2 = 4x \text{ and } x^2 + (y + 12)^2 = 1$$



Since both the tangents at P and Q are parallel.

So their slopes are same.

$$-\frac{2}{y} = -\frac{x}{(y + 12)}$$

$$\Rightarrow \frac{2}{y} = \frac{x}{(y + 12)}$$

$$\Rightarrow 2y + 24 = xy$$

$$\Rightarrow 2y + 24 = y \cdot \frac{y^2}{4}$$

$$\Rightarrow y^3 - 8y - 96 = 0$$

$$\Rightarrow y = 4$$

when $y = 4$, then $x = 4$

So the point Q is (4, 4)

Here, C = (0, -12)

Thus, Shortest distance

$$= PQ$$

$$= CQ - CP$$

$$= \sqrt{16 + 64} - 1$$

$$= \sqrt{80} - 1$$

$$= 2\sqrt{5} - 1$$

55. Given $f(x) = ax + \frac{b}{x}$, $a, b, x > 0$

Applying A.M $\geq$ G.M, we get,

$$\Rightarrow \frac{ax + \frac{b}{x}}{2} \geq \sqrt{ax \cdot \frac{b}{x}} = \sqrt{ab}$$

$$\Rightarrow ax + \frac{b}{x} \geq 2\sqrt{ab}$$

Thus, the min value of $f(x)$ is $2\sqrt{ab}$

56. We have, $f(x) = x^2 + \frac{1}{x^2 + 1}$

$$\Rightarrow f(x) = \left(x^2 + 1 + \frac{1}{x^2 + 1} \right) - 1$$

$$\Rightarrow f(x) \geq 2 - 1 = 1$$

Hence, the least value of $f(x)$ is 1.

57. We have, $f(x) = \frac{x^3 + x + 2}{x}, x > 0$

$$\Rightarrow f(x) = x^2 + \frac{2}{x} + 1$$

$$\Rightarrow f(x) = \left(x^2 + \frac{1}{x} + \frac{1}{x} \right)$$

$$\Rightarrow f(x) \geq 3 + 1 = 4$$

Hence, the min value of $f(x)$ is 4.

58. We have $f(x) = 2\cos x + \sec^2 x$

$$\Rightarrow f(x) = \cos x + \cos x + \sec^2 x$$

$$\Rightarrow f(x) = \cos x + \cos x + \sec^2 x \geq 3$$

Hence, the min value of $f(x)$ is 3.

59. We have $f(x) = 2\log_{10} x - \log_x(0.01), x > 1$

$$\Rightarrow f(x) = 2\log_{10} x - \log_x(10)^{-2}$$

$$\Rightarrow f(x) = 2\log_{10} x + 2\log_x(10)$$

$$\Rightarrow f(x) = 2(\log_{10} x + \log_x(10))$$

$$\Rightarrow f(x) = 2(\log_{10} x + \log_x(10)) \geq 4$$

Hence, the min value of $f(x)$ is 4.

60. We have, $f(x) = \frac{a^2}{\sin^2 x} + \frac{b^2}{\cos^2 x}$

$$\Rightarrow f(x) = \frac{a^2}{\sin^2 x} + \frac{b^2}{\cos^2 x}$$

$$\Rightarrow f(x) = a^2 \operatorname{cosec}^2 x + b^2 \sec^2 x$$

$$\Rightarrow f(x) = a^2(1 + \cos^2 x) + b^2(1 + \tan^2 x)$$

$$\Rightarrow f(x) = a^2 + b^2 + (a^2 \cot^2 x + b^2 \tan^2 x)$$

$$\Rightarrow f(x) \geq a^2 + b^2 + 2ab = (a + b)^2$$

Hence, the min value of $f(x)$ is $(a + b)^2$

61. We have

$$f(x) = 2^x + 3^x + 5^x + \frac{1}{2^x} + \frac{1}{3^x} + \frac{1}{5^x}, x > 0$$

$$\Rightarrow f(x) = \left(2^x + \frac{1}{2^x} \right) + \left(3^x + \frac{1}{3^x} \right) + \left(5^x + \frac{1}{5^x} \right)$$

$$\Rightarrow f(x) \geq 2 + 2 + 2 = 6$$

Hence, the min value of $f(x)$ is 6.

62. We have

$$f(a) = a^{-5} + a^{-4} + 3a^{-3} + \text{Maxima \& Minima}$$

$$\Rightarrow f(a) = a^{10} + a^8 + 1 + \frac{1}{a^5} + \frac{1}{a^4} + \frac{1}{a^3} + \frac{1}{a^3} + \frac{1}{a^3}$$

$$\Rightarrow f(a) = \left(a^{10} + a^8 + \frac{1}{a^5} + \frac{1}{a^4} + \frac{1}{a^3} + \frac{1}{a^3} + \frac{1}{a^3} \right) + 1$$

$$\Rightarrow f(x) \geq 7 + 1 = 8$$

Hence, the min value of $f(a)$ is 8.

63. Given $f(x) = x^{10} + x^7 + \frac{2}{x^3} + \frac{4}{x^2} + \frac{3}{x}, x > 0$

$$\Rightarrow f(x) = x^{10} + x^7 + \left(\frac{1}{x^3} + \frac{1}{x^3} \right)$$

$$+ \left(\frac{1}{x^2} + \frac{1}{x^2} + \frac{1}{x^2} + \frac{1}{x^2} \right) + \left(\frac{1}{x} + \frac{1}{x} + \frac{1}{x} \right)$$

$$\Rightarrow f(x) \geq 11$$

Min value of $f(x)$ is 11.

64. Given

$$f(a, b, c, d) = \frac{(a^2 + 1)(b^2 + 1)(c^2 + 1)(d^2 + 1)}{abcd}$$

$$= \left(\frac{a^2 + 1}{a} \right) \left(\frac{b^2 + 1}{b} \right) \left(\frac{c^2 + 1}{c} \right) \left(\frac{d^2 + 1}{d} \right)$$

$$= \left(a + \frac{1}{a} \right) \left(b + \frac{1}{b} \right) \left(c + \frac{1}{c} \right) \left(d + \frac{1}{d} \right)$$

$$\geq 2.2.2.2 = 16$$

Hence, the min value of $f(a, b, c, d)$ is 16.

65. We have $2x + 3y = 5$

$$\Rightarrow (x + x + y + y + y) = 5$$

$$\text{Now, } \left(\frac{x + x + y + y + y}{5} \right) \geq \sqrt{x^2 \cdot y^3}$$

$$\Rightarrow \left(\frac{5}{5} \right) \geq \sqrt{x^2 \cdot y^3}$$

$$\Rightarrow x^2 \cdot y^3 \leq 1$$

Hence, the max value of $x^2 \cdot y^3$ is 1.

66. We have, $3x + 2y + z = 14$

$$\text{Now, } \left(\frac{x + x + x + y + y + z}{6} \right) \geq \sqrt[6]{x^3 y^2 z}$$

$$\Rightarrow \left(\frac{3x + 2y + z}{6} \right) \geq \sqrt[6]{x^3 y^2 z}$$

$$\Rightarrow \left(\frac{14}{6} \right) \geq \sqrt[6]{x^3 y^2 z}$$

$$\Rightarrow x^3 y^2 z \leq \left(\frac{14}{6}\right)^6 = \left(\frac{7}{3}\right)^6$$

Hence, the max value of $x^3 y^2 z$ is $\left(\frac{7}{3}\right)^6$

67. We have

$$\frac{\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}}{3} \geq \sqrt[3]{\frac{x^2}{a^2} \cdot \frac{y^2}{b^2} \cdot \frac{z^2}{c^2}}$$

$$\Rightarrow \frac{1}{3} \geq \sqrt[3]{\frac{x^2}{a^2} \cdot \frac{y^2}{b^2} \cdot \frac{z^2}{c^2}}$$

$$\Rightarrow \left(\frac{1}{3}\right)^3 \geq \left(\frac{x^2}{a^2} \cdot \frac{y^2}{b^2} \cdot \frac{z^2}{c^2}\right)$$

$$\Rightarrow \left(\frac{1}{3}\right)^3 \geq \left(\frac{xyz}{abc}\right)^2$$

$$\Rightarrow (xyz) \leq \left(\frac{abc}{3^{3/2}}\right)$$

Hence, the max value of xyz is $\left(\frac{abc}{3^{3/2}}\right)$

68. We have $y = \frac{x}{ax^2 + b}$, $a, b, x > 0$

$$\Rightarrow y = \frac{1}{ax + \frac{b}{x}}$$

$$\Rightarrow y \leq \frac{1}{2\sqrt{ab}}$$

Hence, the max value of y is $\frac{1}{2\sqrt{ab}}$

69. We have

$$\begin{aligned} f(x) &= x^2 + 1 + \frac{4}{x^2 + 3} \\ &= (x^2 + 3) + \frac{4}{(x^2 + 3)} - 2 \geq 4 - \end{aligned}$$

Hence, the minimum value of $f(x)$ is 2

70. We have

$$\begin{aligned} f(x, y, z) &= \frac{(x^3 + 2)(x^3 + 2)(x^3 + 2)}{xyz} \\ &= \left(\frac{x^3 + 2}{x}\right) \left(\frac{y^3 + 2}{y}\right) \left(\frac{z^3 + 2}{z}\right) \\ &= \left(x^2 + \frac{2}{x}\right) \left(y^2 + \frac{2}{y}\right) \left(z^2 + \frac{2}{z}\right) \\ &\geq 3.3.3 = 27 \end{aligned}$$

Hence, the minimum value of $f(x)$ is 27.